

XENDE RIVERO BOWERS

PhD Student

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Skills

Mathematical Computing

SageMath, PARI/GP, LaTeX

Programming

Python, C, C++, C#, Java, JavaScript, R, Haskell, SQL

Tools

Git, Jupyter, Visual Studio, Overleaf

Game Development

Unity and Unreal Engine, including Blueprints

Research

Durham University, currently at KCL as a Visiting PGR Student

Working thesis title: *Computing Higher Elliptic Units and the Arguments of Stark Units*

Supervisors: Prof. Herbert Gangl and Dr. Jack Shotton (Durham University), Prof. Owen Patashnick (KCL)

Research: Algebraic number theory, with emphasis on explicit and computational methods: elliptic units and their higher-rank analogues, Stark's conjectures, and explicit class field theory. Focusing on the Bergeron–Charollois–García construction of higher elliptic units for ATR fields, both extending the computational reach of the construction and using it to make conjectural reciprocity laws explicit.

Software: Higher elliptic units in SageMath and PARI/GP. The Bergeron–Charollois–García construction implemented and extended to ATR fields of degree three to seven. Reproduces examples across the cubic, quartic, and quintic cases.

Talks given: *Using the Elliptic Gamma Function to find Abelian Extensions of Cubic Fields*, PGR Research Day (Durham University, 2026)

Conferences attended: HDP Research Day (Bristol University, 2023, 2024, 2025), Y-RANT (Oxford University, 2024)

Experience

Postgraduate Teaching Assistant, Durham University

10/2022 – Present

- Graded coursework and summative assessments for core undergraduate first- and second-year modules, including Complex Analysis, Algebra and Probability.
- Provided detailed feedback to students to improve academic performance and research.

Fact Checker, Numberphile

10/2022 – 10/2023

- Reviewed pre-release drafts of Numberphile videos, checking for mathematical accuracy, as part of the Numberphile Society team.
- Identified and flagged errors, suggested corrections, and helped ensure that mathematical content was correct and clearly presented for a large general audience.

Computer Science Demonstrator, Durham University

10/2021 – 06/2023

- Ran second-year practical laboratory sessions over two academic years, supporting students one-to-one through programming exercises.
- Diagnosed and debugged student code within the classroom, and explained unfamiliar language features and code.

Mathematics Representative for Natural Sciences,

Durham University

10/2021 – 09/2022

Liaised with students and faculty; collected and escalated academic feedback.

Library Stock Manager, St. Cuthbert's Society

10/2021 – 09/2022

Ran lending operations, inventory and procurement.

Education

PhD in Mathematics (Final Year), Durham University

09/2023 – Present

Funded by the Heilbronn Doctoral Partnership (Heilbronn Institute for Mathematical Research)

MSci Natural Sciences, Durham University

09/2019 – 06/2023

Joint degree in Mathematics and Computer Science (First-Class Honours)

Dissertation: *Siegel Modular Forms and Cryptography*

Key Modules: Number Theory, Cryptography & Codes, Topics in Algebra & Geometry, Representation Theory

DipLCM In Classical Guitar, London College of Music

11/2018

Awarded Distinction

A Levels, Homeschooled

06/2018

Maths (A*), Further Maths (A), Physics (A)

Honours & Awards

Outstanding Achievement in Natural Sciences,

Durham University

Awarded in 2019 and 2020

Associateship of King's College - Dean's Commendation,

KCL

Awarded for Semester 1 and Semester 2 in 2026

References

Available on request